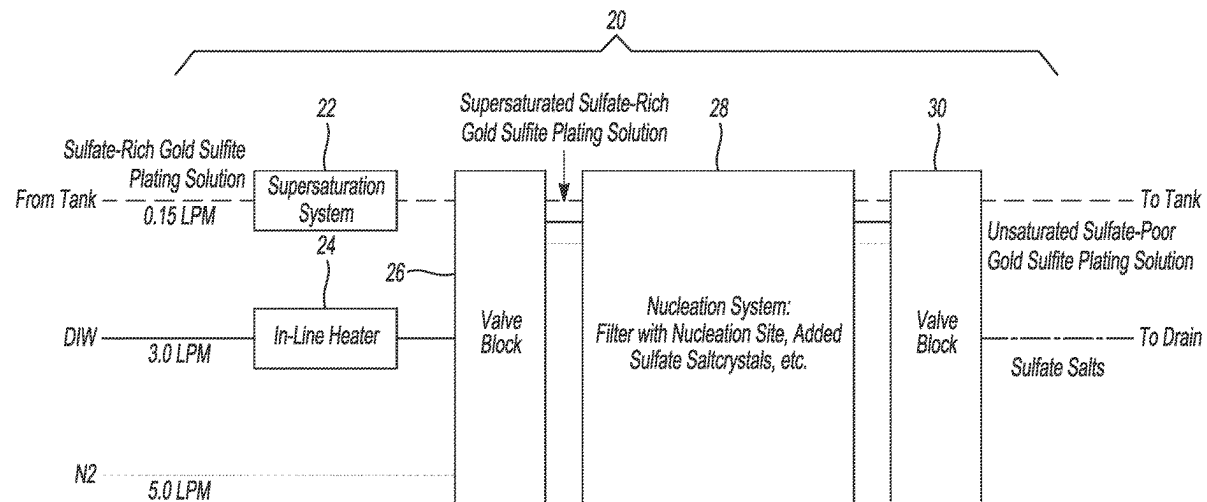




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(19) **United States**(12) **Patent Application Publication**
Bates et al.(10) **Pub. No.: US 2025/0283244 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SULFATE SALT DESALINATION FOR GOLD
SULFITE PLATING SOLUTION****Publication Classification**(71) Applicant: **ClassOne Technology**, Kalispell, MT
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(52) **U.S. Cl.**
CPC **C25D 21/18** (2013.01); **C25D 3/48**
(2013.01)(57) **ABSTRACT**

Sulfate salts are removed from a gold sulfite plating solution via supersaturating a sulfate-rich gold sulfite plating solution to form a sulfate-poor saturated gold sulfite plating solution, and un-saturating the sulfate-poor saturated gold sulfite plating solution to form an unsaturated sulfate-poor gold sulfite plating solution. Gold from the unsaturated sulfate-poor gold sulfite plating solution is electrodeposited onto a workpiece.

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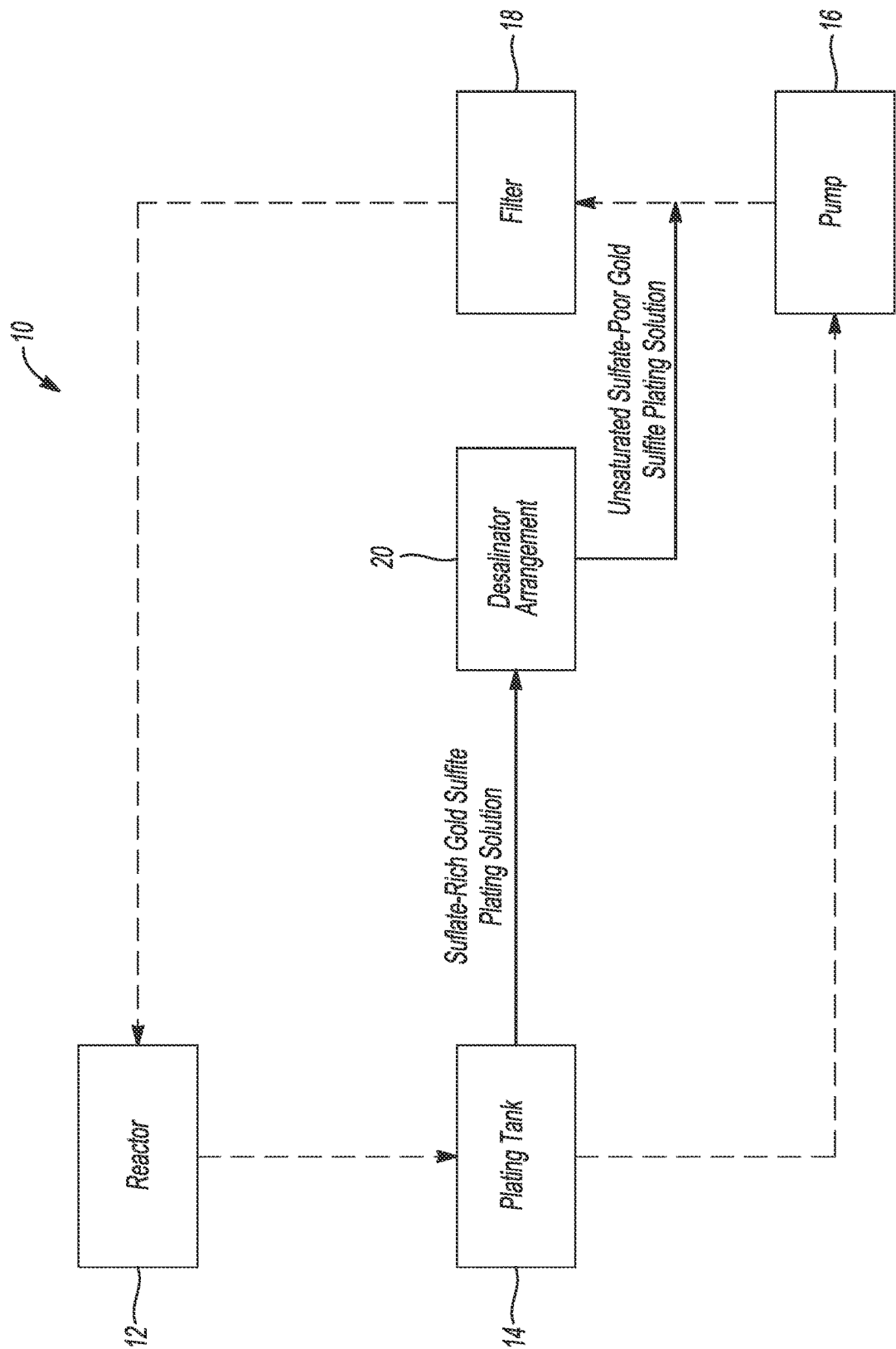


Fig-1

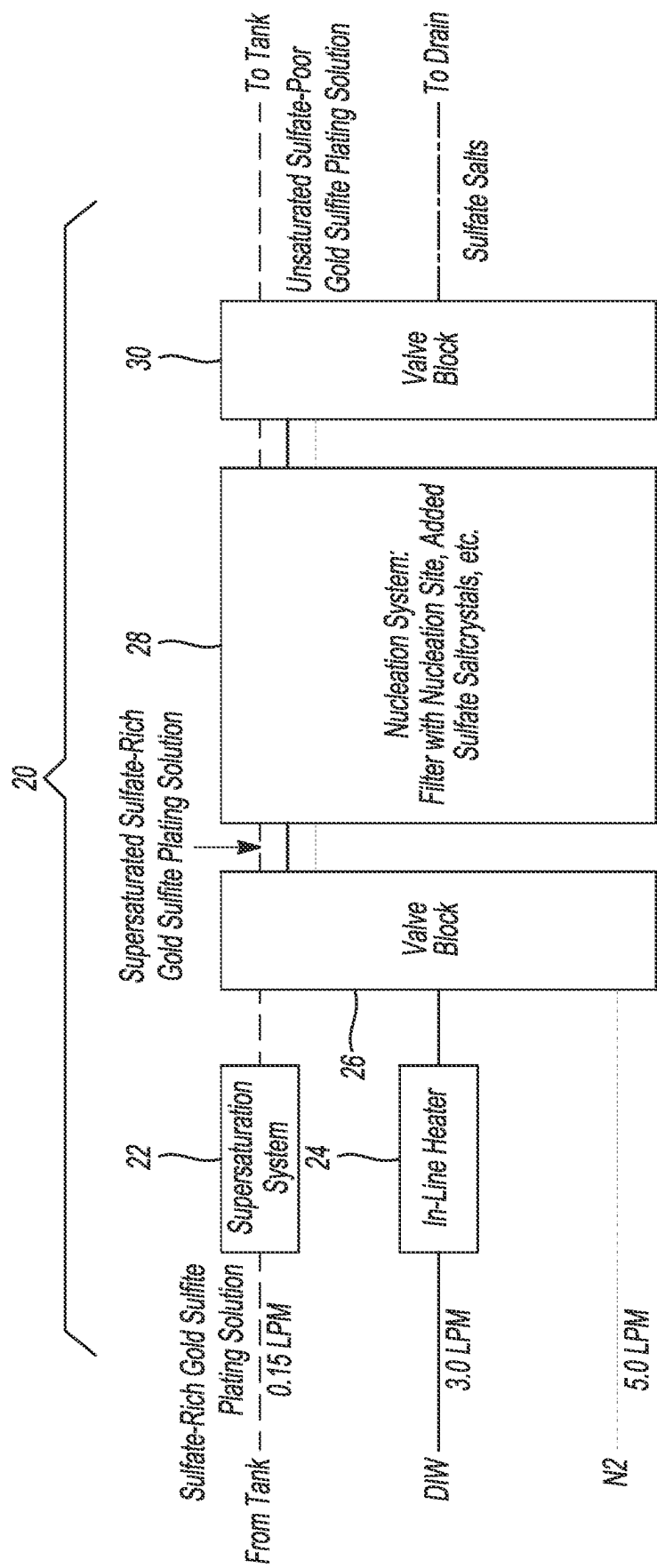


Fig-2

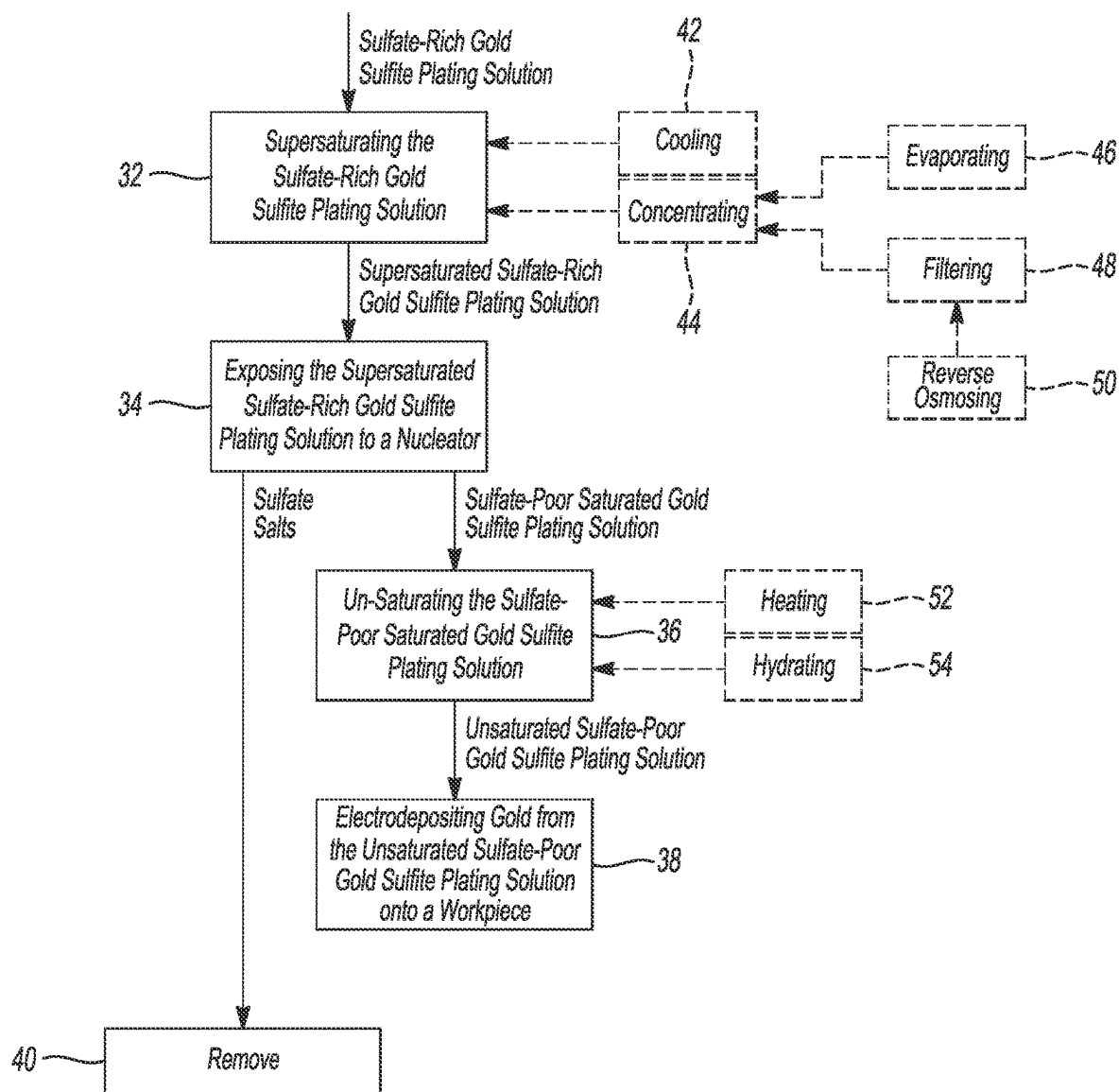


Fig-3

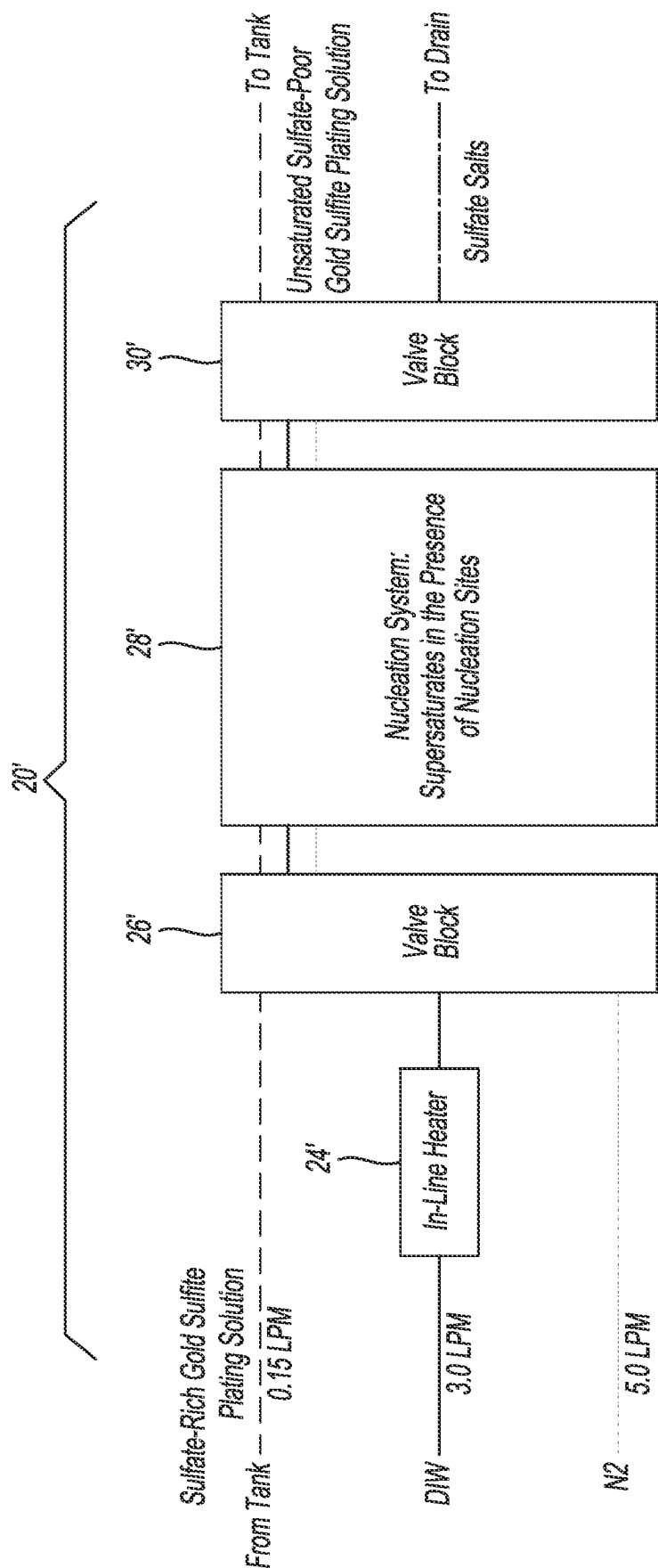


Fig-4

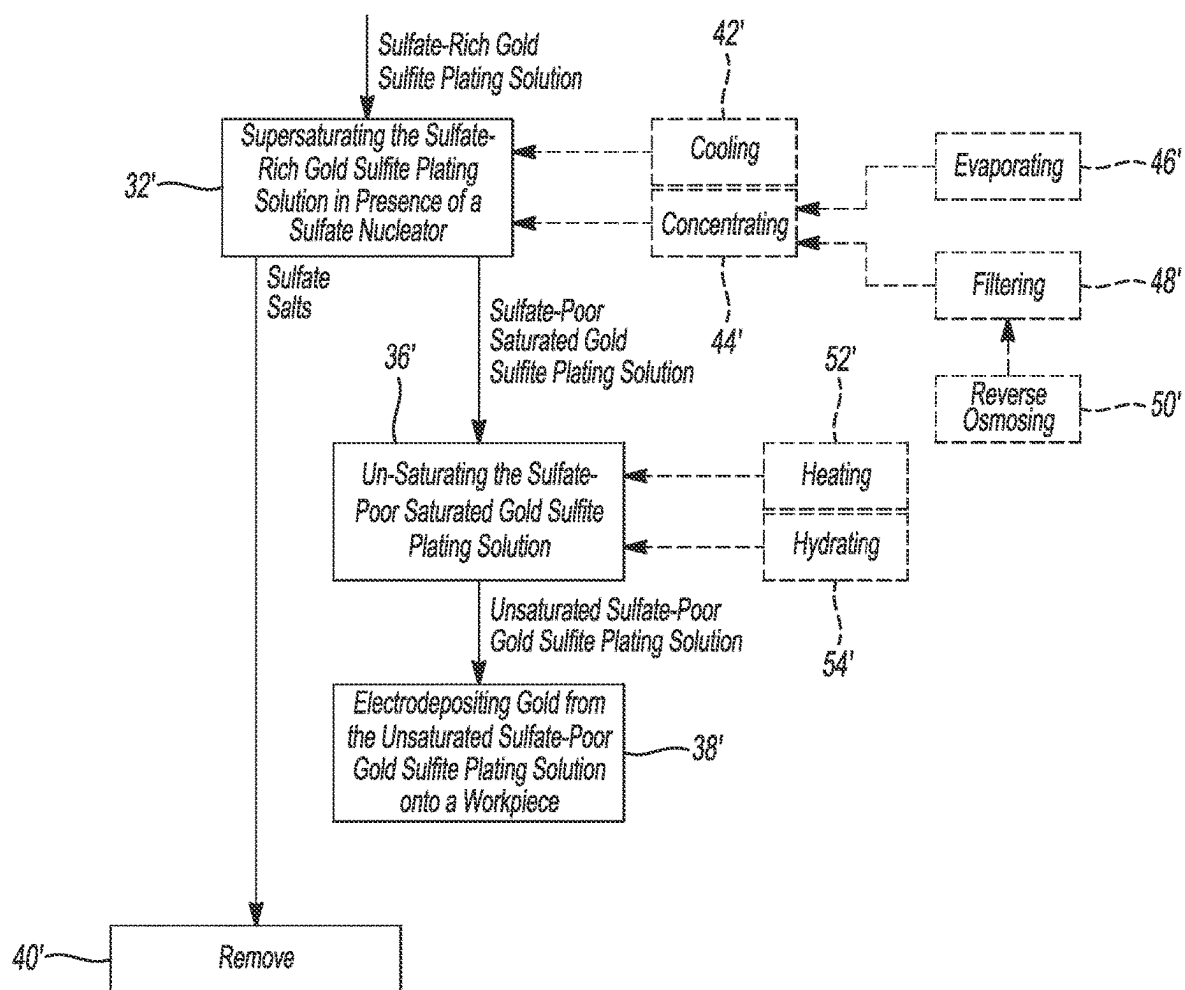


Fig-5

SULFATE SALT DESALINATION FOR GOLD SULFITE PLATING SOLUTION

TECHNICAL FIELD

[0001] This disclosure relates to electrochemical deposition manufacturing equipment.

BACKGROUND

[0002] Electrochemical deposition, a technique in semiconductor fabrication, involves depositing materials onto silicon wafers to create films or patterns. Initially, the wafer's surface is cleansed for optimal adhesion. During deposition, the wafer acts as a cathode in an electrochemical cell with an electrolyte containing the target material's ions. Applying an electrical potential, metal ions reduce and bond to the wafer, forming a layer. This process is controlled by variables like current density and time. Subsequent steps might include annealing and patterning for enhanced quality and specific structures.

SUMMARY

[0003] A method for removing sulfate salts from a gold sulfite plating solution includes supersaturating a sulfate-rich gold sulfite plating solution to form a supersaturated sulfate-rich gold sulfite plating solution, exposing the supersaturated sulfate-rich gold sulfite plating solution to a nucleator to form sulfate salts and a sulfate-poor saturated gold sulfite plating solution, and un-saturating the sulfate-poor saturated gold sulfite plating solution to form an unsaturated sulfate-poor gold sulfite plating solution.

[0004] A method for removing sulfate salts from a gold sulfite plating solution includes supersaturating a sulfate-rich gold sulfite plating solution in a presence of a sulfate nucleator to form sulfate salt precipitate and a sulfate-poor saturated gold sulfite plating solution, and un-saturating the sulfate-poor saturated gold sulfite plating solution to form an unsaturated sulfate-poor gold sulfite plating solution.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a block diagram of an electrodeposition system.

[0006] FIG. 2 is a block diagram of a sulfate salt desalinators system for the electrodeposition system of FIG. 1.

[0007] FIG. 3 is a flow chart of an algorithm for the sulfate salt desalinators system of FIG. 2.

[0008] FIG. 4 is a block diagram of another sulfate salt desalinators system for the electrodeposition system of FIG. 1.

[0009] FIG. 5 is a flow chart of an algorithm for the sulfate salt desalinators system of FIG. 4.

DETAILED DESCRIPTION

[0010] Embodiments are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments may take various and alternative forms. The figures are not necessarily to scale. Some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art.

[0011] Various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

[0012] Gold electroplating can be used for semiconductor applications, particularly metalized silicon wafers. Some may use non-cyanide, mild alkaline compositions. The chemical reactions may involve the reduction of gold ions to elemental gold on the wafer's surface. In such a mild alkaline non-cyanide process, gold ions in the plating solution are reduced to solid gold at the cathode, which is the silicon wafer. The wafer is immersed in the electrolyte solution containing gold ions. When a voltage is applied, the gold ions gain electrons (i.e., are reduced) and deposit as a thin, uniform layer of gold on the wafer. Here, systems and processes to regenerate gold sulfite baths, that use anionic membrane technology, without substantial gold loss at increased concentrations are described.

[0013] Referring to FIG. 1, an electrodeposition system 10 includes a reactor 12, a plating tank 14, a pump 16, and a filter 18. The reactor 12, plating tank 14, pump 16, and filter 18 form a fluid loop such that fluid held by the plating tank 14 is propelled via the pump 16 through the filter 18 and to the reactor 12, with the fluid then returning to the plating tank 14. The electrodeposition system 10 also includes a desalinator arrangement 20. The plating tank 14, pump 16, and desalinator arrangement 20 form a fluid loop such that fluid held by the plating tank 14 is propelled via the pump 16 through the desalinator arrangement 20, with the fluid then returning to the plating tank 14. As discussed in further detail below, the fluid entering the desalinator arrangement 20 in this example is sulfate-rich gold sulfite plating solution, and the fluid exiting the desalinator arrangement 20 is unsaturated sulfate-poor gold sulfite plating solution, which has less sulfate than the sulfate-rich gold sulfite plating solution.

[0014] Referring to FIG. 2, the desalinator arrangement 20 includes a supersaturation system 22, an in-line heater 24, a valve block 26, a nucleation system 28, and a valve block 30. The super saturation system 22 receives sulfate-rich gold sulfite plating solution from the tank 14, and supersaturates the same to produce supersaturated sulfate-rich gold sulfite plating solution. The nucleation system 28, provided the valve 26 is appropriately positioned, receives the supersaturated sulfate-rich gold sulfite plating solution, exposes the same to a nucleator (e.g., crystalized sulfate salt or crystalline structure nucleation sites) to produce sulfate-poor saturated gold sulfite plating solution and sulfate salts, and un-saturates the sulfate-poor saturated gold sulfite plating solution to produce unsaturated sulfate-poor gold sulfite plating solution. The tank 14, provided the valve 30 is appropriately positioned, receives the unsaturated sulfate-poor gold sulfite plating solution.

[0015] To remove the sulfate salts, recirculation of fluid from the tank 14 is stopped. The nucleation system 28, provided the valve 26 is appropriately positioned, receives nitrogen to purge the nucleation system 28. The in-line heater 24 then receives deionized water, and selectively heats the same. That is, the in-line heater 24 may first simply

pass the deionized water, without heating the same, before heating the deionized water to produce heated deionized water. The nucleation system 28, provided the valve is appropriately positioned, receives un-heated deionized water, and then receives heated deionized water to dissolve the sulfate salts to reclaim gold, and provided the valve 30 is appropriately positioned, drains the same. The nucleation system 28, provided the valve is appropriately positioned, then again receives nitrogen to purge the nucleation system 28. Recirculation of fluid from the tank 14 is then started.

[0016] Referring to FIG. 3, at operation 32, sulfate-rich gold sulfite plating solution is supersaturated to produce supersaturated sulfate-rich gold sulfite plating solution. At operation 34, the supersaturated sulfate-rich gold sulfite plating solution is exposed to a nucleator to produce sulfate salts and sulfate-poor saturated gold sulfite plating solution. At operation 36, the sulfate-poor saturated gold sulfite plating solution is un-saturated to produce un-saturated sulfate-poor gold sulfite plating solution. At operation 38, gold from the un-saturated sulfate-poor gold sulfite plating solution is electrodeposited onto a workpiece. At operation 40, sulfate salts are removed.

[0017] Referring again to operation 32, the supersaturating may include cooling the sulfate-rich gold sulfite plating solution at operation 42 or concentrating the sulfate-rich gold sulfite plating solution at operation 44. The concentrating may include evaporating the sulfate-rich gold sulfite plating solution at operation 46 or filtering the sulfate-rich gold sulfite plating solution at operation 48. The filtering may include reverse osmosing the sulfate-rich gold sulfite plating solution at operation 50.

[0018] Referring again to operation 36, the un-saturating may include heating the sulfate-poor saturated gold sulfite plating solution at operation 52 or hydrating the sulfate-poor saturated gold sulfite plating solution at operation 54.

[0019] Referring to FIG. 4, the desalinator arrangement 20' includes an in-line heater 24', a valve block 26', a nucleation system 28', and a valve block 30'. The nucleation system 28', provided the valve 26' is appropriately positioned, receives sulfate-rich gold sulfite plating solution from the tank 14, supersaturates the same in presence of a sulfate nucleator to produce sulfate salts and sulfate-poor saturated gold sulfite plating solution, and un-saturates the sulfate-poor saturated gold sulfite plating solution to produce unsaturated sulfate-poor gold sulfite plating solution. The tank 14, provided the valve 30' is appropriately positioned, receives the unsaturated sulfate-poor gold sulfite plating solution.

[0020] To remove the sulfate salts, recirculation of fluid from the tank 14 is stopped. The nucleation system 28', provided the valve 26' is appropriately positioned, receives nitrogen to purge the nucleation system 28'. The in-line heater 24' then receives deionized water, and selectively heats the same. The nucleation system 28', provided the valve is appropriately positioned, receives un-heated deionized water, and then receives heated deionized water to dissolve the sulfate salts to reclaim gold, and provided the valve 30' is appropriately positioned, drains the same. The nucleation system 28', provided the valve is appropriately positioned, then again receives nitrogen to purge the nucleation system 28'. Recirculation of fluid from the tank 14 is then started.

[0021] Referring again to operation 32', the supersaturating may include cooling the sulfate-rich gold sulfite plating

solution at operation 42' or concentrating the sulfate-rich gold sulfite plating solution at operation 44'. The concentrating may include evaporating the sulfate-rich gold sulfite plating solution at operation 46' or filtering the sulfate-rich gold sulfite plating solution at operation 48'. The filtering may include reverse osmosing the sulfate-rich gold sulfite plating solution at operation 50'.

[0022] Referring again to operation 36', the un-saturating may include heating the sulfate-poor saturated gold sulfite plating solution at operation 52' or hydrating the sulfate-poor saturated gold sulfite plating solution at operation 54'.

[0023] Table 1 shows example concentrations of gold, sulfite, and sulfate, and density for an example plating solution before and after application of the desalination techniques contemplated herein:

TABLE 1

	Before	After	Comment
Gold (g/l)	12.65	12.7	Gold not removed
Sulfite (g/l)	33.8	34.84	Sulfite not removed
Sulfate (g/l)	104.9	69.36	Sulfate reduced
Density	20.74	16.12	Density reduced

[0024] The algorithms, methods, or processes disclosed herein can be deliverable to or implemented by a computer, controller, or processing device, which can include any dedicated electronic control unit or programmable electronic control unit. Similarly, the algorithms, methods, or processes can be stored as data and instructions executable by a computer or controller in many forms including, but not limited to, information permanently stored on non-writable storage media such as read only memory devices and information alterably stored on writeable storage media such as compact discs, random access memory devices, or other magnetic and optical media. The algorithms, methods, or processes can also be implemented in software executable objects. Alternatively, the algorithms, methods, or processes can be embodied in whole or in part using suitable hardware components, such as application specific integrated circuits, field-programmable gate arrays, state machines, or other hardware components or devices, or a combination of firm-ware, hardware, and software components.

[0025] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of these disclosed materials.

[0026] As previously described, the features of various embodiments may be combined to form further embodiments of the invention that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics may be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes may include, but are not limited to strength, durability, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, embodiments described as

less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and may be desirable for particular applications.

What is claimed is:

1. A method for removing sulfate salts from a gold sulfite plating solution, comprising:

supersaturating a sulfate-rich gold sulfite plating solution to form a supersaturated sulfate-rich gold sulfite plating solution;

exposing the supersaturated sulfate-rich gold sulfite plating solution to a nucleator to form sulfate salts and a sulfate-poor saturated gold sulfite plating solution; and un-saturating the sulfate-poor saturated gold sulfite plating solution to form an unsaturated sulfate-poor gold sulfite plating solution.

2. The method of claim 1 further comprising electrodepositing gold from the unsaturated sulfate-poor gold sulfite plating solution onto a workpiece.

3. The method of claim 1, wherein the supersaturating includes cooling the sulfate-rich gold sulfite plating solution.

4. The method of claim 1, wherein the supersaturating includes concentrating the sulfate-rich gold sulfite plating solution.

5. The method of claim 4, wherein the concentrating includes evaporating.

6. The method of claim 4, wherein the concentrating includes filtering.

7. The method of claim 6, wherein the filtering is via reverse osmosis.

8. The method of claim 1, wherein the un-saturating includes heating the sulfate-poor saturated gold sulfite plating solution.

9. The method of claim 1, wherein the un-saturating includes hydrating the sulfate-poor saturated gold sulfite plating solution.

10. The method of claim 1, wherein the nucleator is crystalized sulfate salt or crystalline structure nucleation sites.

11. A method for removing sulfate salts from a gold sulfite plating solution, comprising:

supersaturating a sulfate-rich gold sulfite plating solution in a presence of a sulfate nucleator to form sulfate salt precipitate and a sulfate-poor saturated gold sulfite plating solution; and

un-saturating the sulfate-poor saturated gold sulfite plating solution to form an unsaturated sulfate-poor gold sulfite plating solution.

12. The method of claim 11 further comprising electrodepositing gold from the unsaturated sulfate-poor gold sulfite plating solution onto a workpiece.

13. The method of claim 11, wherein the supersaturating includes cooling the sulfate-rich gold sulfite plating solution.

14. The method of claim 11, wherein the supersaturating includes concentrating the sulfate-rich gold sulfite plating solution.

15. The method of claim 14, wherein the concentrating includes evaporating.

16. The method of claim 14, wherein the concentrating includes filtering.

17. The method of claim 16, wherein the filtering is via reverse osmosis.

18. The method of claim 11, wherein the un-saturating includes heating the sulfate-poor saturated gold sulfite plating solution.

19. The method of claim 11, wherein the un-saturating includes hydrating the sulfate-poor saturated gold sulfite plating solution.

20. The method of claim 11, wherein the sulfate nucleator is crystalized sulfate salt or crystalline structure nucleation sites.

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